

Asymptotic Analysis



```
def is_prime(n)
    div_count = 0 #1
    for curr in range(1, n+1):
        if n % curr == 0: #1
            div_count += 1 #1
    return div_count == 2 #1
```

```
def print_squares(n)
    for i in range(1, n+1):
        line = 'x' * n
        print(line) #1
```

line = n
for i in range(n):
line += 'x'

$$T(n) = \sum_0^n (n) + 1 = \Theta(n^2)$$

$$T(n) = 1 + \sum_0^n (1+1) + 1 = 2n + 2 \cong \Theta(n)$$

$\Theta \rightarrow$ How the runtime grows as $n \rightarrow \infty$ exactly

$\Theta \rightarrow$ The upper bound (worst-case scenario) as $n \rightarrow \infty$



Big-O Analysis

$f(n)$: runtime of your algo (how many steps as n grows)

$g(n)$: Some function (like n , n^2 , $\log n$), that we use to compare how fast $f(n)$ grows

$$f(n) = O(g(n))$$

C - constant

n_0 - the point when n is "really big"

$$f(n) \leq C \cdot g(n)$$

$$f(n) = 3n^2 + 6n - 15, \quad g(n) = cn^2$$

$$C = 4$$

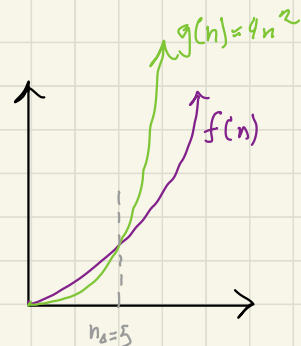
$$g(n) = 4n^2$$

$$3(5)^2 + 6(5) - 15 \leq 4(5)^2$$

$$75 + 30 - 15 \leq 100$$

$$90 \leq 100$$

$$3n^2 + 6n - 15 \leq 4n^2$$



Correctness ✓

Ω - How fast the algo is guaranteed to grow ("best-case scenario")

$f(n) \rightarrow$ runtime as $n \rightarrow \infty$

$g(n) \rightarrow$ function to lower-bound $f(n)$

$$f(n) = \Omega(g(n))$$

$$f(n) \geq c \cdot g(n)$$

$$f(n) = 3n^2 + 6n - 15$$

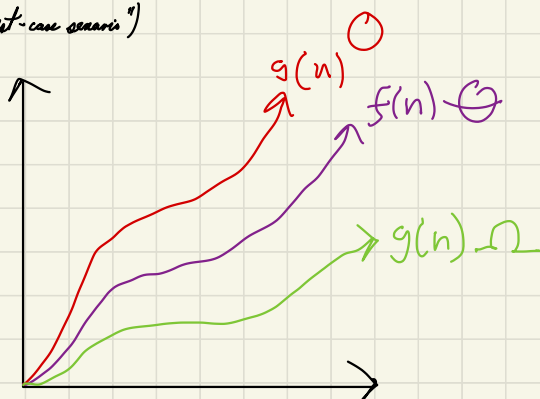
$$g(n) = cn^2 \rightarrow c = 2.5$$

$$3n^2 + 6n - 15 \geq 2.5n^2$$

$$3(5)^2 + 6(5) - 15 \geq 2.5(5)^2$$

$$90 \geq 62.5$$

Correctness ✓



$f(n) = \Theta(g(n))$ if there exists values c_1, c_2 & n_0 such that

$$c_1 g(n) \leq f(n) \leq c_2 g(n)$$

$$f(n) = 3n^2 + 6n - 15$$

linear growth insignificant constant

$$c_1 = 3$$

$$c_2 = 9$$

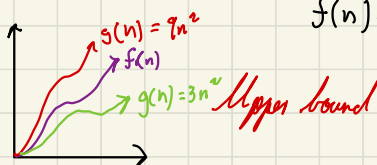
$$n_0 = 3$$

the dominant term

$$3n^2 \leq f(n) \leq 9n^2$$

$$f(n) = \Theta(n^2)$$

Lower Bound



$$3n^2 + 6n - 15 \geq 3n^2$$

$$3(3)^2 + 6(3) - 15 \geq 3(3)^2$$

$$27 + 18 - 15 \geq 27$$

Correctness ✓

$$3n^2 + 6n - 15 \leq 9n^2$$

$$3(3)^2 + 6(3) - 15 \leq 81$$

$$27 + 18 - 15 \leq 81$$

Correctness ✓ $30 \leq 81$